ML Assignment 1

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1 Exercise 1: - Probabilities (Theoretical)

Solve the following problems:

a. In a population of n couples, where each couple has exactly 2 kids. What is the probability that all couples have 1 daughter and 1 son? (assume binary equiprobable genders)

In order to answer this question, I have to take into account the following:

$$P(\text{boy}) = P(\text{girl}) = \frac{1}{2}$$

The events $P(\text{boy})$ and $P(\text{girl})$ are independent. To get one boy and one girl, I have to either get a boy in the first birth and a girl in the second, or the opposite.

So the probability is give by multipying n times the sum of the two possibilities:

Since both probabilities are equal to $\frac{1}{2}$:

$$P(\text{one boy and one girl}) = \left(\frac{1}{4} + \frac{1}{4}\right)^n = \left(\frac{1}{2}\right)^n$$

b. You flip a fair (50-50 chance of each side) coin n times. What is the probability that all n times you get Heads?

The probability of getting heads is

$$\frac{1}{2}$$

So to get that n consecutive times the probability is

$$\left(\frac{1}{2}\right)^n$$

c. In a box, there are beads of 3 colours: red, green, and blue at percentages 3050hollow. Which is the probability of drawing a hollow bead by picking a random bead from the box? The conditional probabilities (hollow — color) are given so i have to marginalize for the color The answer is

$$\frac{1}{2} * \frac{3}{10} + \frac{2}{3} * \frac{5}{10} + \frac{2}{3} * \frac{2}{10} =$$

2 Exercise 2: - Bayesian Theorem (Programming)

a. A photon package emitted by a star has a probability of 1e-7 to pass through the Earth's atmosphere and reach a detector placed on the ground. The detector has the following properties:

i) A false positive rate (= probability of incorrectly reporting the detection of a photon package, when no photon package was actually received) of 10

ii) A true positive rate (= probability of correctly reporting the detection of a photon package, when a photon package was actually received) of 85

Which is the probability that if the detector reported a detection - a photon package was actually received?

To answer this I have to use the Bayes theorem The prior probability of the photon is 1e-7

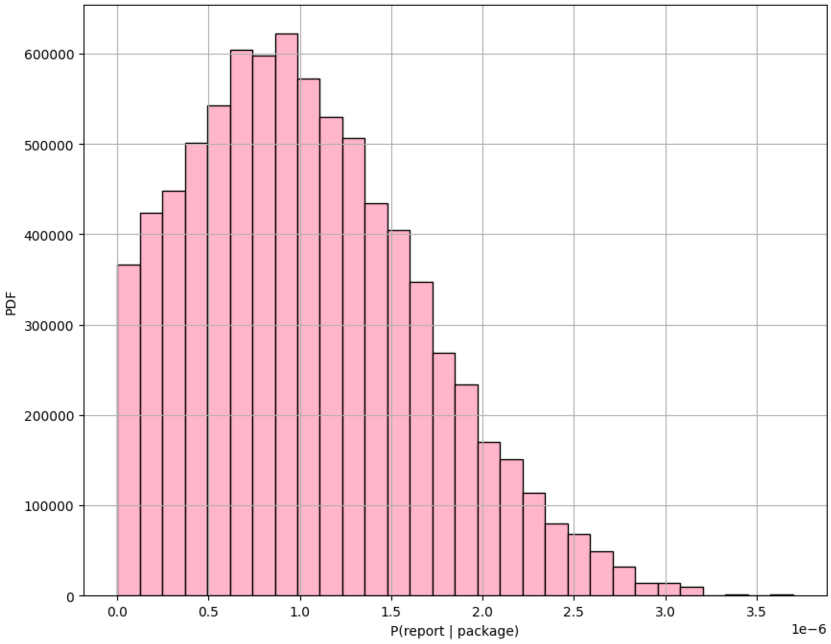
The true positives can be used as the conditional probability of reporting the photon given that the photon exists

The false positive can be used as the conditional probability of reporting a photon given that the photon do not exists.

The marginal probability of reporting a photon is the sum of rerporting the photon when it exists ad reporting the photon when it do not exists.

Plugging these numbers in the Bayes theoreem formula I get a probability of 8.49999362500478e-07

b. Assume that the photon package is composed of 100 photons. Each photon has an equal probability [i.e. 25%] to carry an energy of either 10, 20, 30, or 40 electronvolts. That means that each package has photon energies roughly distributed according to a uniform distribution in energy. Which

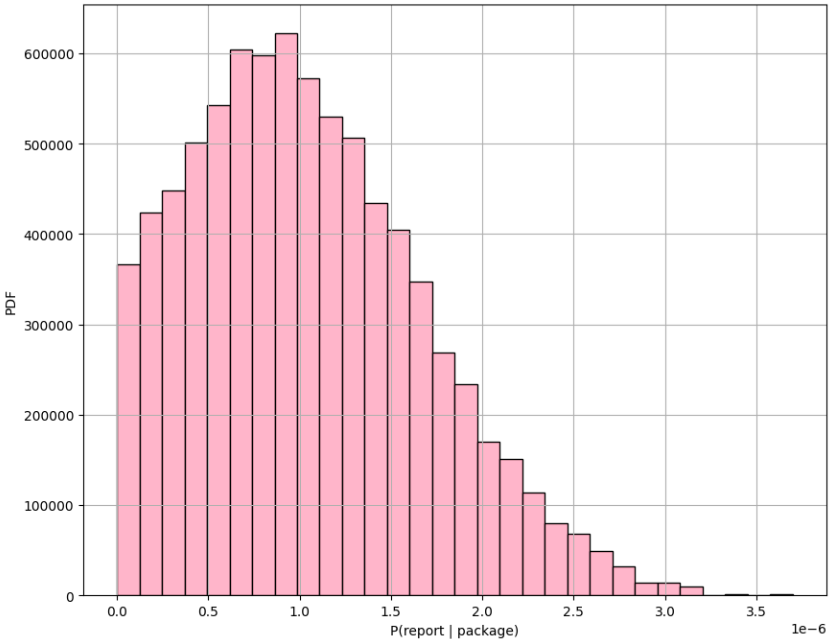


distribution the total energy of a package (i.e. the sum of the energies of the photons composing a package) follows? Plot it. Does it look like a familiar distribution? Why? To answer this question I can generate a defined number of packages(here 2000) based on the probabilities of each photon. The total energy of each package should be a number deviating around the expected mean which is: the sum of the probability of the photon multiplied by its energy. The histogram resembles a normal distribution around the expected value.

c. We will now assume that the photon package doesn't simply have a defined probability of 1e-7 to reach the ground. Instead, due to stochastic emission from the star and interaction with the atmosphere, the probability density function for this event is described by a normal distribution N (μ,) centered at μ = 1e-7, and with = 9e-8. The resulting probability that if the detector reported a detection a photon package was actually received will now be a distribution (instead of a single value). Plot such distribution. Does it make sense?

Following the same procedure as in the first part of the exercise but this time I sample to get the probability of the photon I sample from a gaussian with mean 1e-7 and standard deviation 9e-8.

I compute the conditional probabilities of reporting a detection given that a package has been received using the bayes theorem. Note that beacause probabilities have to be positive number and the given mean is close to zero



I have to exclude the negative probabilities that have been sampled